

Lab Assignment – 9

Decision Trees

Question 1: Learning Axis-Aligned vs Oblique Boundaries

Objective: To explore a known limitation of Decision Trees — their tendency to form axis-aligned splits, and how this affects their ability to learn more complex decision boundaries.

Generate a dataset of 800 samples where both x_1 and x_2 are uniformly distributed in the range $[-2, 2]$. Define a new variable:

$$z = x_1 + x_2 + 0.3 \sin(3x_1)$$

Assign labels such that $y = 1$ if $z > 0$, and $y = 0$ otherwise.

Now, train Decision Tree models with different depths (2, 4, 6, 8, and 10). For each case, plot the decision boundary. You should be able to observe how the model approximates the true boundary as depth increases.

Next, introduce an additional feature:

$$x_3 = x_1 + x_2$$

Retrain the models using this extended feature set and again visualize the decision boundaries.

Finally, compare the results. Try to reason out:

- why the tree struggles initially,
- how the additional feature changes the learning behaviour,
- and whether increasing depth alone is sufficient to capture such boundaries.

Question 2: Decision Tree from Scratch

Objective: The aim is to understand how Decision Trees actually work internally, instead of relying entirely on library implementations.

Implement a basic Decision Tree classifier from scratch. At a minimum, the implementation should include:

- computation of the Gini index,
- a method to find the best split at each node,
- recursive construction of the tree,
- and reasonable stopping conditions (such as maximum depth or minimum samples per node).

You may either use an existing dataset or generate one (around 500 samples should be sufficient).

Once the implementation is complete, train models with depths 3, 5, and 7. Compare your results with a standard library implementation.

While comparing, report the accuracy and explain:

- whether both models behave similarly,
- if there are differences in splits or structure,
- and any limitations in your implementation.

Question 3: Using Decision Tree in a Real-World Dataset

Choose any one dataset from Kaggle (for example, Titanic survival prediction, heart disease dataset, or customer churn). Perform the necessary preprocessing steps such as handling missing values and encoding categorical features.

Train a Decision Tree model and experiment with different hyperparameters like depth and minimum samples per split. Try to tune the model so that it performs reasonably well without overfitting.

Evaluate your model using appropriate metrics (accuracy, F1-score, confusion matrix, etc.), and also look at feature importance.

In your analysis, focus on:

- which features are influencing the decision most,
- whether the model is overfitting,
- and how the results differ from the synthetic datasets used earlier.

Question 4: Noise Robustness

Objective: To explore how sensitive Decision Trees are to noise.

Generate a dataset of 1000 samples using the following relation:

$$z = x_1^2 + x_2^2 - 4$$

Assign labels based on whether $z > 0$ or not.

Now, introduce noise in two ways:

- by randomly flipping a certain percentage of labels (10%, 20%, 30%),
- and by adding noise to the input features.

Train Decision Tree models on these noisy datasets and record the performance.

Try to compare:

- how label noise and feature noise affect the model differently,
- whether the model becomes unstable as noise increases,
- and why Decision Trees might be particularly sensitive in such cases.